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## RESEARCH ARTICLE

# YOLO-LSD: A Lightweight Object Detection Model for Small Targets at Long Distances to Secure Pedestrian Safety

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**ABSTRACT** Ensuring the safety of children, the elderly, and individuals with disabilities remains a significant challenge in modern transportation environments, particularly in urban areas with mixed traffic comprising both vehicles and pedestrians. Real-time detection of long-distance and small-scale objects remains difficult due to resolution and computational constraints. This study proposes an improved lightweight object detection model, You Only Look Once – Long-distance Small-target Detection (YOLO-LSD), designed for the long-range recognition of small objects in intelligent transportation applications. The proposed model integrates the C3C2 and the new Efficient Layer Aggregation Network - Convolutional Block Attention Module (ELAN-CBAM) modules to improve the efficiency of feature extraction while reducing computational overhead. The C3C2 module optimizes the network structure by reducing redundant operations, making it more suitable for real-time deployment on embedded devices. The CBAM module improves feature selection by incorporating channel and spatial attention mechanisms, thereby enhancing the robustness of small-object detection under complex urban conditions. The proposed YOLO-LSD model was tested on a customized wearable backpack system equipped with front and rear cameras for environmental perception. YOLO-LSD attains higher detection accuracy for small and distant objects while maintaining lower computational complexity. This study employs the PASCAL VOC2007+2012 (VOC0712) dataset to evaluate the proposed model. Compared with YOLOv7, the model achieves a maximum mean Average Precision (mAP) of 80.1%, a minimum parameter count of 30.539M, and the lowest computational cost of 91.6 GFLOPS. This lightweight architecture makes it particularly suitable for intelligent transportation, pedestrian safety, and autonomous mobility applications. In summary, this lightweight architecture, YOLO-LSD, is particularly suitable for intelligent transportation, pedestrian safety and automated mobility applications.

**INDEX TERMS** Attention mechanism, lightweight, YOLO, small object detection, feature extraction, convolutional neural networks, convolutional block attention module (CBAM), similarity-based attention mechanism (SimAM).

## I. INTRODUCTION

In modern transportation environments, ensuring the safety of children, the elderly and people with disabilities have become a critical issue. This challenge is particularly evident in urban

areas with mixed traffic, where these vulnerable groups often struggle to quickly perceive potential hazards due to slower reaction times or limited visibility caused by their height restrictions [1], [2]. Deep learning-based object detection techniques have been widely applied in intelligent transportation and pedestrian protection systems. Among them, the You Only Look Once (YOLO) algorithm has demonstrated

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exceptional real-time performance and accuracy [3]. However, the complexity of urban environments, such as vehicle occlusion, lighting changes, and background noise, still pose challenges to object detection algorithms [4].

Beyond detecting nearby objects around people, real-time recognition of distant targets is equally crucial, especially in high-speed vehicle environments where moving objects gradually expand from small to large within the image frame. This phenomenon significantly increases the difficulty of detecting distant objects [5]. In addition, the detection accuracy of small objects is limited by limited resolution, occlusions, and background interference, leading to a high miss rate [6]. These challenges are particularly evident in scenarios of intelligent driving and automated monitoring. For example, errors in traffic sign detection due to insufficient resolution could impact vehicle decision making and compromise pedestrian safety [7].

To address these challenges, various improvement strategies have been proposed, such as Feature Pyramid Networks (FPN) and Channel Attention (CA), which enhance the recognizability of small objects [7]. Additionally, with the rise of edge computing and mobile device applications, model lightweight has become a major research focus in recent years. Architectures such as MobileNet and ShuffleNet have been used to reduce computational costs while ensuring real-time processing capabilities [8]. These technological advances have enabled object detection models to be deployed more effectively in mobile devices, autonomous aerial vehicles (AAVs), and intelligent transportation systems [9]. However, existing technologies still face a trade-off between model lightweighting and detection accuracy. Therefore, the key research challenge remains to maintain high detection performance while minimizing computational costs [10].

In real-time recognition of distant targets, Soft-NMS [4] reduces false alarms and improves IoU loss functions (EIoU, SIoU) to improve the accuracy of the location of the bounding box. Moreover, in long-range detection applications, strengthening feature extraction and integrating contextual information are key to improving recognition capabilities. For example, the Multi-Scale Feature Pyramid Network (MS-FPN) has been demonstrated to significantly improve the recognition rate of small objects in remote sensing imagery [11]. In road monitoring scenarios, feature enhancement techniques based on Swin-Transformer have also been proven effective in improving small-object detection performance [10]. In highly dynamic environments, attention mechanisms integrated with Fixated Object Detection (FOD) can effectively filter key targets, further enhancing the stability of detection systems. These technological advances indicate that future research will focus on addressing the challenges of long-range small object detection by strengthening multiscale learning, reducing computational burden, and improving model adaptability [12].

This study addresses the challenges of detecting road objects for children, the elderly and people with disabilities

in intelligent transportation scenarios. This study propose an improved YOLO-LSD method that integrates multi-scale feature extraction with a lightweight attention mechanism to enhance the detection accuracy of long-range small objects. The main contributions of this study are as follows.

- 1) An improved C3 module (C3C2) is introduced to enhance feature extraction capability. By reducing redundant computations, it improves computational efficiency while minimizing inference latency, making the model more suitable for edge computing devices and real-time applications.
- 2) The Convolutional Block Attention Module (CBAM) is introduced to enhance feature selection. By leveraging channel and spatial attention mechanisms, it improves the recognition capability of critical regions, making it particularly effective for small-object detection and long-range recognition. This approach significantly reduces detection errors caused by background noise or lighting variations.
- 3) The Similarity-based Attention Mechanism (SimAM) is integrated to enhance feature representation capability. SimAM utilizes a neuron energy function computation mechanism to improve feature selectivity without introducing additional parameters or increasing computational complexity. This enables the model to focus more effectively on critical visual information, further enhancing the recognition performance of small objects.
- 4) Implementation of a high-performance lightweight model: By optimizing the network architecture and reducing computational overhead, the proposed model ensures high detection accuracy even in resource-constrained environments such as mobile devices, while also supporting real-time processing requirements.
- 5) Using the CBAM and SimAM mechanisms to enhance the capability of selecting target features, the detection rate of small distant targets (such as pedestrians and traffic signs) is significantly improved, while the false alarm rate is effectively reduced. This makes the approach more suitable for intelligent transportation and autonomous driving scenarios.

## II. RELATED WORKS

### A. ONE-STAGE AND TWO-STAGE OBJECT DETECTION APPROACHES

Object detection techniques can be broadly categorized into one-stage and two-stage methods. One-stage methods, such as the YOLO series, perform classification and bounding box regression in a single forward pass, making them suitable for real-time applications. In [13], YOLOv5s is employed, and by improving the network architecture and incorporating an efficient ECA Attention mechanism, the small object detection capability is enhanced, making it more adaptable to various scenarios. In [14], the YOLOv5 model is integrated with a Vision Transformer (ViT) to improve global feature extraction capability, thus improving detection accuracy while maintaining high-speed inference.

In [15], a Stack-YOLO model designed for foreign object detection is proposed. This model optimizes the network architecture through the CSS (Cross, Split, Shuffle) design, effectively reducing computational complexity while enhancing object detection accuracy. In [16], a model based on YOLOv5s is proposed. By integrating a lightweight transformer and the Coordinate and Adaptive Pooling Attention (CAPA) attention mechanism, the model enhances small object detection capabilities, making it suitable for AAV inspection and intelligent transportation applications. In comparison, two-stage methods such as faster R-CNN [17] utilize a region proposal network (RPN) to select potential target regions. Although these methods generally achieve higher accuracy, they typically suffer from a slower inference speed [17]. In [18], various trainable release quota methods were proposed, which can significantly improve detection accuracy in real-time object detection without increasing inference costs. In [19], a network scaling method was introduced to maintain an optimal balance between speed and accuracy.

### B. ATTENTION MECHANISMS FOR SMALL OBJECT DETECTION

Small object detection in the field of computer vision faces challenges in feature extraction. To address this issue, researchers have widely applied attention mechanisms to enhance detection performance. In [20], the Efficient Multiscale Attention (EMA) mechanism was employed to strengthen cross-channel and cross-spatial feature fusion, thereby improving the accuracy of small object detection in AAV imagery. In [21], a small object recognition algorithm based on hybrid control and feature fusion was proposed. This method enhances small object feature extraction capabilities through Relative Position Encoding. In [22], a multi-scale feature fusion network was designed to address the challenge of small object detection in AAV aerial imagery, enabling more precise detection of small targets in AAV images. Additionally, a study proposed an optimized YOLOv5 model for object detection in aerial imagery. This approach integrates the CBAM to enhance small object feature extraction and introduces a Feature Enhancement Module (FEM) to improve detection capabilities across different scales. The proposed method demonstrated outstanding performance on the VisDrone and AAV123 datasets [23]. To enhance small object detection in low-light scenarios, a study proposed the use of an Adaptive Residual Feature Block (ARFB) to strengthen image features. Additionally, an adaptive enhancement module was incorporated to improve brightness adjustment and texture enhancement, allowing effective target detection in low-light environments [24]. In [25], an improved version of YOLOv5s was introduced for power inspection. This approach utilizes the Efficient Channel Attention Network (ECA-Net) mechanism, allowing for more accurate identification of distant or small foreign objects.

### C. ROAD OBJECT DETECTION AND DEPLOYMENT DEVICES

Road object detection is a critical component of intelligent transportation and autonomous driving technologies. In [26], MCS-YOLO was proposed, incorporating Swin Transformer to enhance feature extraction capabilities and integrating a multi-scale small object detection structure to improve the accuracy of detecting dense small objects. In [27], a real-time traffic sign and road object detection method was introduced, designed for deployment on various edge computing platforms such as Jetson Xavier AGX, Jetson Xavier NX, Jetson Nano, and test PCs. By optimizing the network structure, this method enables efficient operation on resource-constrained devices, making it suitable for edge computing environments. For monitoring-based traffic detection, a small object detector is proposed, YOLO SOD [28], specifically designed for urban mixed traffic scenarios. This model integrates a Super-Resolution Feature Extraction Module (SRFEM) and Knowledge Distillation techniques to enhance small object detection performance in complex environments. To address the challenge of multi-object tracking in complex traffic scenarios, IMANet (Interactively Motion-Assisted Network) was proposed in [29]. This network enhances multi-object tracking accuracy by incorporating a motion-assisted module and leveraging feature matching and temporal sequence information to reduce target loss and misidentification risks. Additionally, in [30], a tiling-based small object detection technique was employed to improve the recognition capability of small objects from a AAV perspective.

## III. GUIDELINES FOR MANUSCRIPT PREPARATION

### A. PURPOSE MODEL STRUCTURE

The study proposes an improved lightweight object detection model, YOLO-LSD, with the model architecture illustrated in Fig. 1. The backbone network has been modified to incorporate the proposed ELAN-CBAM module, which integrates the Efficient Layer Aggregation Network (ELAN) and the CBAM. Additionally, the C3C2 module is introduced in the neck to enhance detection accuracy while reducing computational costs. This architecture optimizes feature extraction and multi-scale feature fusion, making it suitable for real-time object detection applications.

#### 1) BACKBONE IMPROVEMENTS

The backbone of YOLO-LSD adopts the ELAN-CBAM module to replace the original ELAN module in YOLOv7, as illustrated in the A1–A4 blocks of Fig. 1, to enhance feature extraction efficiency and representation capability. The primary objective of this improvement is to accelerate feature learning by leveraging the dense residual connections ResNet-like structures and concatenation mechanism in the Efficient Layer Aggregation Network (ELAN), ensuring that gradient flow is maintained throughout the deep feature extraction process. Additionally, the CBAM is employed to enhance feature selectivity, emphasizing critical regions while suppressing irrelevant background information.

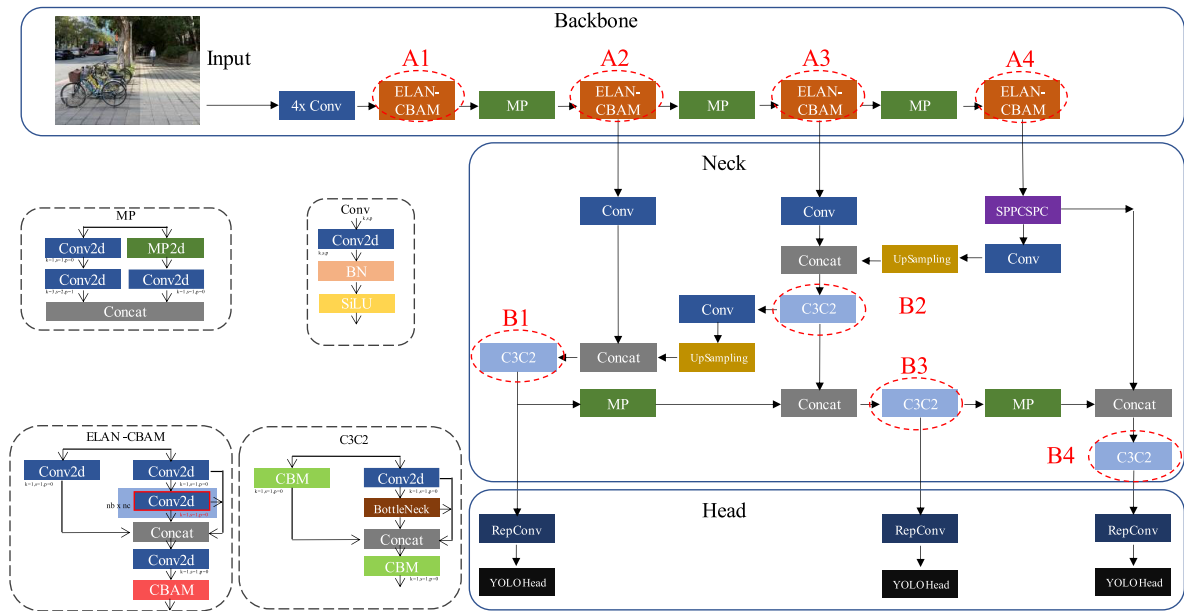


FIGURE 1. The network structure of YOLO-LSD.

The channel attention mechanism (CAM) strengthens the response to key features of the target object while mitigating the influence of background noise. Furthermore, the spatial attention mechanism (SAM) enables the model to focus on important regions within the image, allowing the backbone to prioritize target object locations rather than uniformly processing the entire image.

## 2) NECK IMPROVEMENTS

In the standard neck structure of YOLOv7, ELAN is primarily used for feature fusion and transmission. However, in the YOLO-LSD model, this study found that the neck had a negligible effect on feature map extraction. Therefore, all ELAN modules were replaced with C3C2 (corresponding to B1–B4 in Fig. 1) to reduce computational costs and enhance feature representation capability. Compared to the original ELAN in the YOLOv7 neck, the C3C2 module incorporates an optimized bottleneck design, effectively reducing computational demands, making it suitable for real-time applications. C3C2 retains the deep feature aggregation capability inherent in the YOLO series while ensuring that critical information is preserved during the concatenation process. Furthermore, by employing residual learning, the model maintains stable gradient flow as depth increases, preventing the problem of vanishing gradient during training.

### B. CBAM: CONVOLUTIONAL BLOCK ATTENTION MODULE

CBAM is a lightweight and versatile attention mechanism that can be seamlessly integrated into any convolutional neural network (CNN) to enhance feature representation capability. This module combines channel attention and spatial attention, sequentially refining the input features, as illustrated in Fig. 2 [1], [31].

CBAM first applies the channel attention mechanism to determine the importance of different channels in the feature map. This process extracts global feature information using global average pooling and global max pooling, which are then processed through a shared fully connected layer to compute channel-wise weights. Finally, the sigmoid function is applied to activate the calculated weights and generate the channel attention map. The channel attention mechanism processes the input feature map  $F \in \mathbb{R}^{C \times H \times W}$  to generate the spatial weight map  $M_S(F') \in \mathbb{R}^{C \times 1 \times 1}$ . After applying the calculated channel weights, the final feature map can be obtained as shown in (1) [1], [31].

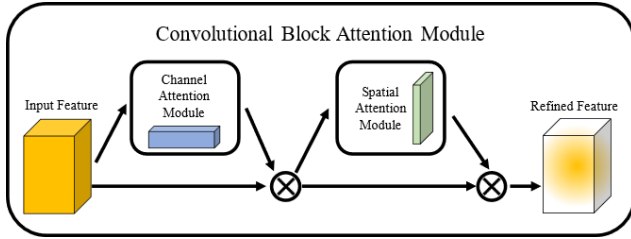
$$F' = M_c(F) \otimes F, \quad (1)$$

After computing the channel-wise attention weights, CBAM further employs spatial attention to highlight critical spatial regions. The spatial attention mechanism first applies max pooling and average pooling to the channel-compressed feature map. The pooled features are then processed through a convolution operation to generate spatial attention weights. Finally, the sigmoid activation function is applied to produce the spatial attention map  $M_S(F') \in \mathbb{R}^{1 \times H \times W}$ . According to (2) [1], [31], the output feature map after applying spatial attention to the weighted features is obtained.

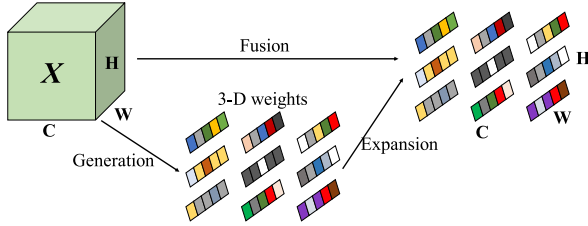
$$F'' = M_s(F') \otimes F', \quad (2)$$

CBAM features low computational overhead and broad applicability, allowing flexible integration into various CNN architectures without disrupting the original training process. Additionally, CBAM can be trained together with the primary network without requiring additional supervisory signals. In tasks such as object detection, image classification, and semantic segmentation, CBAM demonstrates exceptional





**FIGURE 2.** Structure of the CBAM, including channel attention module and spatial attention module structure.



**FIGURE 3.** Structure of the SimAM attention mechanism, where the attention module incorporates complete 3D weights.

feature enhancement capabilities. Notably, it significantly improves recognition accuracy in small object detection.

By jointly learning channel and spatial attention, CBAM effectively strengthens critical features while suppressing unnecessary background information. This enables the model to focus more on valuable visual features, thereby enhancing overall performance and generalization capability.

### C. SIMAM: SIMILARITY-BASED ATTENTION MECHANISM

Similarity-based Attention Mechanism (SimAM) is a lightweight attention mechanism designed based on neuroscience principles, proposed by Liang et al. in 2021. Unlike traditional channels or spatial attention mechanisms, SimAM uses an energy function that does not require additional parameters to assess the importance of neurons, thus enhancing the selectivity and representation capability of the network's feature.

The design of SimAM is inspired by the concept of information-rich neurons in visual neuroscience. These neurons exhibit distinct firing patterns compared to their neighboring neurons and often demonstrate strong spatial inhibition, which plays a crucial role in visual processing tasks. SimAM determines the importance of a target neuron in the feature map by assessing its linear separability from adjacent neurons. This approach effectively models neuron interactions and generates a 3D weight distribution suitable for high-dimensional features, as illustrated in Fig. 3.

In the SimAM mechanism, each pixel in the feature map is treated as an independent neuron, and its significance is computed using an energy function. For each target neuron, the energy function is defined as shown in (3) [32].

$$e_t = \frac{\frac{4}{M} \sum_{i=1}^M (x_i - a)^2 + 4\beta}{(t - a)^2 + \frac{2}{M} \sum_{i=1}^M (x_i - a)^2 + 2\beta}, a = \frac{1}{M} \sum_{i=1}^M x_i \quad (3)$$

Here, the input feature is represented by  $H$  and  $W$  for the spatial dimensions, while  $C$  denotes the number of channels.

$M$  represents the number of neurons within a channel,  $\alpha$  is the average value of all neurons within the channel excluding the target neuron, and  $\beta$  is a fixed adjustment parameter, typically set to a very small value. Based on the computed energy function, the weight of each neuron can be expressed as shown in (4) [32].

$$\bar{X} = \text{sigmoid}\left(\frac{1}{E}\right) \odot X \quad (4)$$

Here,  $E$  represents the sum of values across all channels and spatial locations. The  $\text{sigmoid}(\cdot)$  function is used to constrain the weight range, preventing excessively large values from affecting model convergence. The symbol  $\odot$  denotes element-wise multiplication (Hadamard product).  $X$  represents the original input feature map, and  $(X)^{-}$  denotes the feature map refined by SimAM.

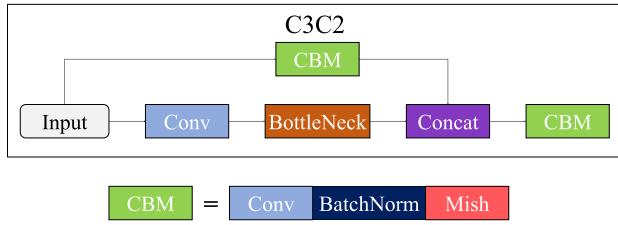
The most notable characteristic of SimAM is that it does not require additional learnable parameters, achieving significant enhancements of features at minimal computational cost. Compared to traditional attention mechanisms, SimAM preserves both channel and spatial information while avoiding increased model complexity, making it particularly suitable for resource-constrained applications. Furthermore, SimAM has demonstrated outstanding performance in object detection, image classification, and semantic segmentation tasks. In particular, it effectively improves recognition capability in small object detection tasks, where fine-grained feature extraction is critical.

This energy function-based weight computation not only aligns with the principles of biological visual mechanisms but also provides a more efficient attention selection strategy for existing deep learning models.

### D. C3C2

The C3C2 module is an improved feature extraction and fusion mechanism designed to enhance computational efficiency in object detection networks, playing a crucial role in the YOLOv7 architecture. This module is optimized based on the Cross Stage Partial (CSP) structure, addressing the redundant feature aggregation issues that may arise in the traditional C3 module. Compared to the standard C3, C3C2 reduces unnecessary convolution operations and concatenation processes, making the computation more efficient. This optimization not only improves feature extraction capabilities, allowing the model to better integrate spatial and semantic information, but also reduces inference latency. Additionally, C3C2 employs the Mish activation function, further enhancing gradient propagation and network stability, enabling the model to maintain high performance while remaining lightweight, making it suitable for real-time applications [33].

In the conventional YOLOv7 architecture, ELAN involves extensive convolutional layers (Conv2d) and feature concatenation operations, significantly increasing computational



**FIGURE 4.** Structure of the C3C2, including improved CSP module and Mish activation function.

costs and memory consumption, negatively affecting real-time inference efficiency. This issue is particularly pronounced when the model is deployed on edge computing devices, where computational resources are limited. To address these challenges, this study replaces the ELAN structure in the neck with the C3C2 module, specifically corresponding to B1/C1, B2/C2, B3/C3, and B4/C4, as illustrated in Fig. 1. C3C2 streamlines the computation process by reducing unnecessary convolution and concatenation operations while preserving essential feature learning capabilities, thereby improving computational efficiency. Furthermore, C3C2 enhances feature extraction and fusion, allowing the model to reduce inference latency and memory usage without compromising accuracy, making it more suitable for real-time object detection applications, as shown in Fig. 4.

## IV. EXPERIMENT

### A. EXPERIMENT SETUP

The PASCAL VOC dataset is utilized in this study, which consists of VOC2007 and VOC2012. The training set is constructed by integrating all data from VOC2012 with the training set of VOC2017. Validation is performed on VOC2017val, and the final testing is conducted on VOC2017test. The experiments are conducted using an NVIDIA RTX 4090.

This study evaluates the model performance using three metrics: Precision, Recall, and mean Average Precision (mAP). Precision (P) represents the ratio of correctly identified positive samples to the total number of detected samples, measuring how many of the detected samples are true positives. The corresponding formula is given in (5) [32].

$$P = \frac{TP}{TP + FP} \quad (5)$$

Recall (R) reflects the proportion of correctly detected positive samples among all actual positive samples, which measures the model's ability to comprehensively identify target objects. The corresponding formula is given in (6) [32].

$$R = \frac{TP}{TP + FN} \quad (6)$$

mAP is a comprehensive evaluation metric that calculates the average precision at different threshold values to assess the overall accuracy of the model. The corresponding

formulas are given in Equations (7) and (8) [32].

$$AP = \sum_{k=1}^N P(k) \Delta r(k) \quad (7)$$

$$mAP = \frac{\sum_{i=1}^n AP_i}{n} \quad (8)$$

In the definition of correct and incorrect classifications, TP (True Positive) and TN (True Negative) represent the correctly identified positive and negative samples, respectively. FP (False Positive) occurs when the model incorrectly classifies a negative sample as positive, leading to an increased False Alarm Probability. FN (False Negative) indicates that the model fails to detect an actual positive sample, raising the Missed Detection Probability.

In practical applications, FP may result in false alarms, where the model mistakenly detects a defect in a non-defective area, while FN may cause missed detections, preventing the model from identifying actual defects. For instance, in PCB inspection, FP can generate unnecessary alerts, whereas FN may result in undetected defects, compromising the reliability of the inspection system.

The proposed model was tested on a custom-designed backpack-mounted device. The backpack device was designed on the basis of a typical backpack used in daily life. Considering the detection of objects approaching from both the front and rear, a front-facing camera was installed to detect objects approaching from the front, while a rear-facing camera was added to identify objects gradually approaching from behind. A real-world photograph of the device is shown in Fig. 5.

As shown in Fig. 6, the system adopts Jetson Orin NX as the core computing unit, responsible for processing visual data from the front and rear cameras while executing real-time object detection and recognition algorithms to provide accurate environmental perception. The device is equipped with two cameras to enhance the user's awareness of approaching objects in the environment.

To ensure system stability, the computing module is connected to a Power Management IC (PMIC), which Dynamically regulates power distribution to meet energy demands under different operating conditions. The power system supports two power sources: Power input from an external power supply and a battery, ensuring adaptability to various application scenarios. This design enables stable operation across diverse environments, thereby enhancing walking safety and assisting functionality.

### B. EXPERIMENT RESULTS

This study focuses on optimizing YOLOv7, aiming to reduce the number of parameters and computational complexity while maintaining high detection accuracy to accommodate resource-constrained embedded systems and real-time applications. The experiments are implemented using the PyTorch framework with CUDA acceleration, systematically evaluating attention mechanisms, lightweight structures, and

TABLE 1. Impact of attention mechanisms on YOLO-LSD.

| NO.            | A1   | A2  | A3   | A4   | C1    | C2    | C3    | C4    | mAP   | Precision | Recall | Parameter | GFLOPS |
|----------------|------|-----|------|------|-------|-------|-------|-------|-------|-----------|--------|-----------|--------|
| YOLOv7         | N/A  | N/A | N/A  | N/A  | N/A   | N/A   | N/A   | N/A   | 77.8% | 76.8%     | 71.9%  | 36584258  | 103.5  |
| YOLOv7-SimAM-1 | N/A  | N/A | N/A  | N/A  | SimAM | SimAM | SimAM | SimAM | 79.7% | 78.5%     | 73.7%  | 36890750  | 104.0  |
| YOLOv7-CBAM-1  | CBAM | N/A | N/A  | N/A  | N/A   | N/A   | N/A   | N/A   | 78.9% | 78.7%     | 72.4%  | 36592821  | 103.5  |
| YOLOv7-CBAM-2  | N/A  | N/A | CBAM | CBAM | N/A   | N/A   | N/A   | N/A   | 79.2% | 78.7%     | 72.6%  | 56315387  | 103.9  |
| YOLOv7-CBAM-3  | CBAM | N/A | CBAM | CBAM | N/A   | N/A   | N/A   | N/A   | 75.2% | 74.9%     | 69.1%  | 56315387  | 139.6  |

TABLE 2. The impact of lightweight methods on YOLO-LSD.

| NO.           | B1   | B2   | B3   | B4   | mAP   | Precision | Recall | Parameter | GFLOPS |
|---------------|------|------|------|------|-------|-----------|--------|-----------|--------|
| YOLOv7        | N/A  | N/A  | N/A  | N/A  | 77.8% | 76.8%     | 71.9%  | 36584258  | 103.5  |
| YOLOv7-C3C2-1 | C3C2 | N/A  | C3C2 | C3C2 | 78.8% | 75.4%     | 75%    | 35374978  | 97.3   |
| YOLOv7-C3C-1  | C3C2 | C3C2 | N/A  | C3C2 | 78.4% | 76.8%     | 73%    | 31507074  | 94.2   |
| YOLOv7-C3C2-3 | C3C2 | C3C2 | C3C2 | C3C2 | 78.8% | 77.1%     | 73.2%  | 30539778  | 91.1   |

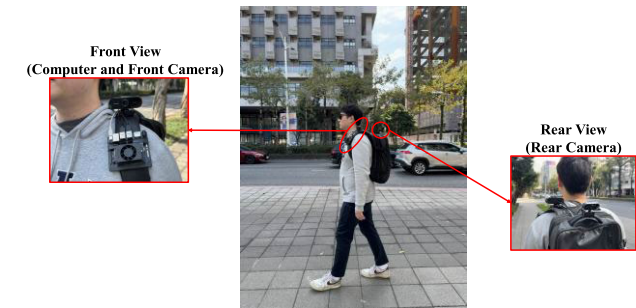


FIGURE 5. Introduction to the proposed custom-designed backpack device.

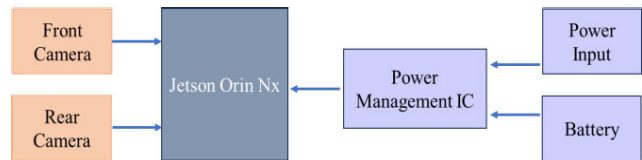


FIGURE 6. Architecture of the proposed custom-designed backpack device.

comprehensive optimization strategies to determine the optimal model configuration.

To analyze the impact of attention mechanisms, CBAM and SimAM were introduced into the Backbone of YOLOv7 for an ablation study, assessing their effects on object detection performance, as shown in Table 1. The CBAM module was incrementally integrated into the backbone, and when the ELAN module was completely replaced with ELAN-CBAM (YOLOv7-CBAM-4), the mAP reached 80.1%,

outperforming the baseline YOLOv7 (77.8%), demonstrating the significant benefits of CBAM in enhancing feature extraction and small object detection.

In contrast, SimAM primarily improved Precision (78.5%) but exhibited a slightly lower Recall than CBAM, indicating its advantage in local feature selection. However, CBAM incurs a higher computational cost than SimAM, with GFLOPS reaching 104.0, suggesting the need for further optimization in resource-limited environments.

In the evaluation of lightweight structures, structural adjustments were made to the YOLOv7 Neck, replacing the ELAN module with C3C2 to reduce the number of parameters and computational burden, as shown in Table 2. The results indicate that when all ELAN modules in the Neck were replaced with C3C2 (YOLOv7-C3C2-3), the mAP reached 78.8%, while the number of parameters significantly decreased to 30.5M, and GFLOPS was reduced to 91.1, demonstrating that C3C2 improves computational efficiency while maintaining reasonable detection performance.

The detection mAP of YOLOv7-C3C2-1 was the same as that of YOLOv7-C3C2-3, but it exhibited higher parameter count and GFLOPS. In contrast, YOLOv7-C3C2-2, which partially replaced ELAN, achieved an mAP of 78.4%, with GFLOPS reduced to 94.2, indicating that moderate lightweight modifications can effectively reduce computational overhead.

By integrating the effects of CBAM and C3C2, this study further optimizes the model to determine the most suitable architecture for real-time applications, as shown in Table 3. Introducing CBAM into the Backbone and replacing ELAN

**TABLE 3.** Impact of integrating attention mechanisms and lightweight methods on YOLO-LSD.

| NO.    | Backbone | Head         | mAP   | Parameter | GFLOPS |
|--------|----------|--------------|-------|-----------|--------|
| YOLOv7 | N/A      | N/A          | 0.778 | 36584258  | 103.5  |
| E1     | N/A      | C3C2         | 0.788 | 30539778  | 91.1   |
| E2     | CBAM     | N/A          | 0.801 | 36890750  | 104.0  |
| E3     | N/A      | SimAM        | 0.797 | 36890750  | 104.0  |
| E4     | CBAM     | C3C2         | 0.792 | 30846270  | 91.6   |
| E5     | N/A      | C3C2 + SimAM | 0.784 | 30539778  | 91.1   |
| E6     | CBAM     | C3C2 + SimAM | 0.788 | 30846270  | 91.6   |

**TABLE 4.** Identification results of commonly encountered objects on the road.

| Class     | YOLOv7 | YOLO-LSD |
|-----------|--------|----------|
| bicycle   | 89.2%  | 89%      |
| bird      | 73.3%  | 72.6%    |
| bottle    | 65%    | 63.7%    |
| bus       | 86.8%  | 88.1%    |
| car       | 90.5%  | 90.9%    |
| cat       | 84.1%  | 84.4%    |
| chair     | 59%    | 61.8%    |
| dog       | 80.7%  | 81%      |
| motorbike | 87.2%  | 89.2%    |
| person    | 86.7%  | 87.4%    |
| train     | 83.6%  | 86.9%    |

with C3C2 in the Neck (E4) increases mAP to 79.2%, reduces the number of parameters to 30.8M, and achieves GFLOPS of 91.6, striking an optimal balance between accuracy and computational efficiency.

Compared to CBAM (E2) alone, applying CBAM (E2) increases mAP to 80.1%, but also increases GFLOPS to 104.0, making it more suitable for high-performance scenarios. However, this configuration may not be ideal in resource-constrained environments. Configurations E5 and E6, which combine C3C2 with SimAM, reduce GFLOPS to 91.1 but achieve lower mAP than E4, highlighting CBAM's stronger impact on recognition accuracy while C3C2 primarily optimizes computational resources.

The proposed YOLOv7-CBAM-C3C2 model reduces computational costs while maintaining recognition performance, making it suitable for real-time computing and resource-constrained applications, as shown in Fig. 3. CBAM enhances feature extraction and improves small object detection, while C3C2 effectively reduces computational overhead. By integrating both mechanisms, the model achieves an optimal balance between accuracy and computational efficiency.

Table 4 presents the recognition results for common objects encountered on the road. Compared to the YOLOv7 model, the proposed method achieves better recognition performance in most detection categories. The model is tested on the PASCAL VOC2007 test set, and the recognition results are compared with those of YOLOv7 by evaluating the AP of each category.

**TABLE 5.** Comprehensive comparison of the proposed model with other YOLO-based and non-YOLO object detection models.

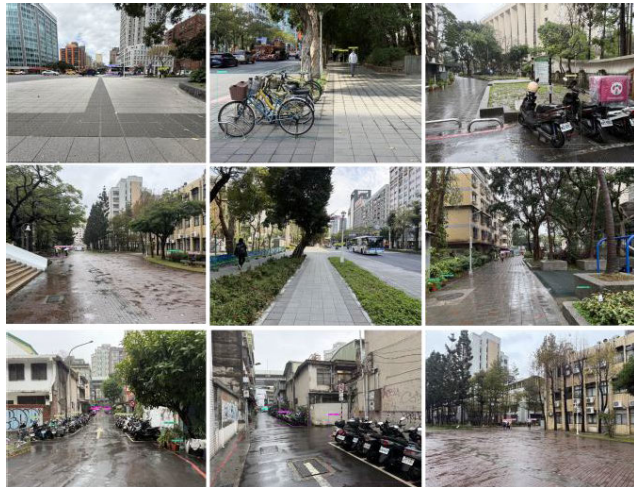
| Class            | mAP          | Parameters     |
|------------------|--------------|----------------|
| Faster R-CNN[T7] | 76.4%        | -              |
| YOLO4-tiny[T8]   | 52.5%        | 59.180M        |
| SSD300[G3]       | 74.3%        | -              |
| Fast R-CNN[G3]   | 70%          | -              |
| YOLOv7-tiny[T6]  | 77.8%        | 60.654M        |
| YOLOv7           | 77.8%        | 36.584M        |
| <b>YOLO-LSD</b>  | <b>79.2%</b> | <b>30.846M</b> |

The results show that the proposed model achieves higher AP values across various recognition categories. Even for chairs, which exhibit the lowest recognition performance, the method significantly improves AP. The proposed approach demonstrates superior capability in multi-class object detection experiments, further validating its applicability.

The proposed model is comprehensively compared with other YOLO-based and non-YOLO object detection models to evaluate its effectiveness, as shown in Table 5. According to the results in Table 5, the proposed model outperforms others in terms of mAP and the number of parameters. Compared to YOLO4-tiny, which has the lowest mAP score, the proposed model achieves a significant improvement of 66.28%.

A visual performance analysis is conducted using images captured by the custom-developed backpack device. The detection results, as shown in Fig. 7, the images obtained from the customized backpack system demonstrate that the proposed model can accurately identify objects, particularly excelling in the precise localization of small objects. The analysis of the visual results indicates that the model can consistently detect various types of targets, including pedestrians, bicycles, motorcycles, buses, and urban infrastructure, while maintaining robust recognition performance under different environmental conditions, such as lighting variations, object occlusions, and background noise. Furthermore, observations from the images show that the model performs well in detecting small and distant objects, accurately drawing bounding boxes and distinguishing adjacent objects, thereby reducing false detections and missed instances. Notably, even in low-contrast and wet ground conditions, object contours remain clearly marked, demonstrating the model's strong adaptability to complex urban environments.





**FIGURE 7.** Images captured by the custom backpack device and the corresponding object detection results of the model.

## V. CONCLUSION

Children, the elderly, and people with physical disabilities often struggle to quickly perceive potential hazardous objects in their surroundings due to visual and attentional limitations, particularly on the streets or in crowded environments. However, there is currently a lack of effective assistive tools capable of real-time object detection and safety alerts. Moreover, existing object detection models are computationally intensive and involve complex parameters, making them unsuitable for resource-constrained embedded systems or mobile devices. This limitation significantly hinders their ability to achieve real-time operation on lightweight hardware.

This study presents an improved version of YOLOv7 by introducing a lightweight object detection model aimed at reducing the number of parameters and computational complexity. The Neck is redesigned using the C3C2 module to minimize redundant computations while maintaining detection accuracy. Additionally, the CBAM module integrates channel and spatial attention mechanisms to enhance small object feature extraction, ensuring effective recognition of potential hazardous objects even in resource-constrained environments.

The experimental results demonstrate that the model integrating CBAM and C3C2 achieves a mAP of 80.1%. Compared to standard YOLOv7, it maintains high accuracy while reducing both the number of parameters and computational complexity. Through C3C2 optimization, the model's parameter count is reduced to 30.5M, and its GFLOPS decreases to 91.1, effectively minimizing model size and computational demands, making it more suitable for embedded devices. Furthermore, ablation studies indicate that applying CBAM to the Backbone and C3C2 to the Head helps reduce computational overhead while preserving detection performance.

The proposed model is designed for resource-constrained mobile and wearable devices, such as smart safety backpacks, smart glasses, and portable navigation devices. It enables

children, the elderly, and people with physical disabilities to quickly recognize both moving and stationary objects in their surroundings, reducing the risk of collisions and accidents. By ensuring a lightweight architecture and low computational resource consumption, the model delivers high-performance object detection, enhancing safety across various environments.

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